Basic Sales Summary using SQLite and Python

N RITHYA REDDY
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Sales Summary using SQL Aggregation

Query Objective:

Summarize sales data by product

Calculate total quantity sold

Compute total revenue per product

SQL Concepts Used:

SUM() for aggregation

GROUP BY to group data by product

Revenue calculated as quantity × price

Result:

Notebook generated highest revenue

Aggregated data successfully using SQL inside Python

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```python
# Main SQL Summary Query
query = """
SELECT
    product,
    SUM(quantity) AS total_quantity,
    SUM(quantity * price) AS revenue
FROM sales
GROUP BY product
"""

# Execute query and load into pandas
df = pd.read_sql_query(query, conn)

print("Sales Summary:")
print(df)
```

```
Sales Summary:
    product  total_quantity  revenue
0  Notebook              15    300.0
1       Pen              15     75.0
2    Pencil              30     90.0
```

Revenue Analysis by Product

Analysis Objective:

Visualize total revenue generated by each product

Compare product performance

Key Observations:

Notebook generated the highest revenue

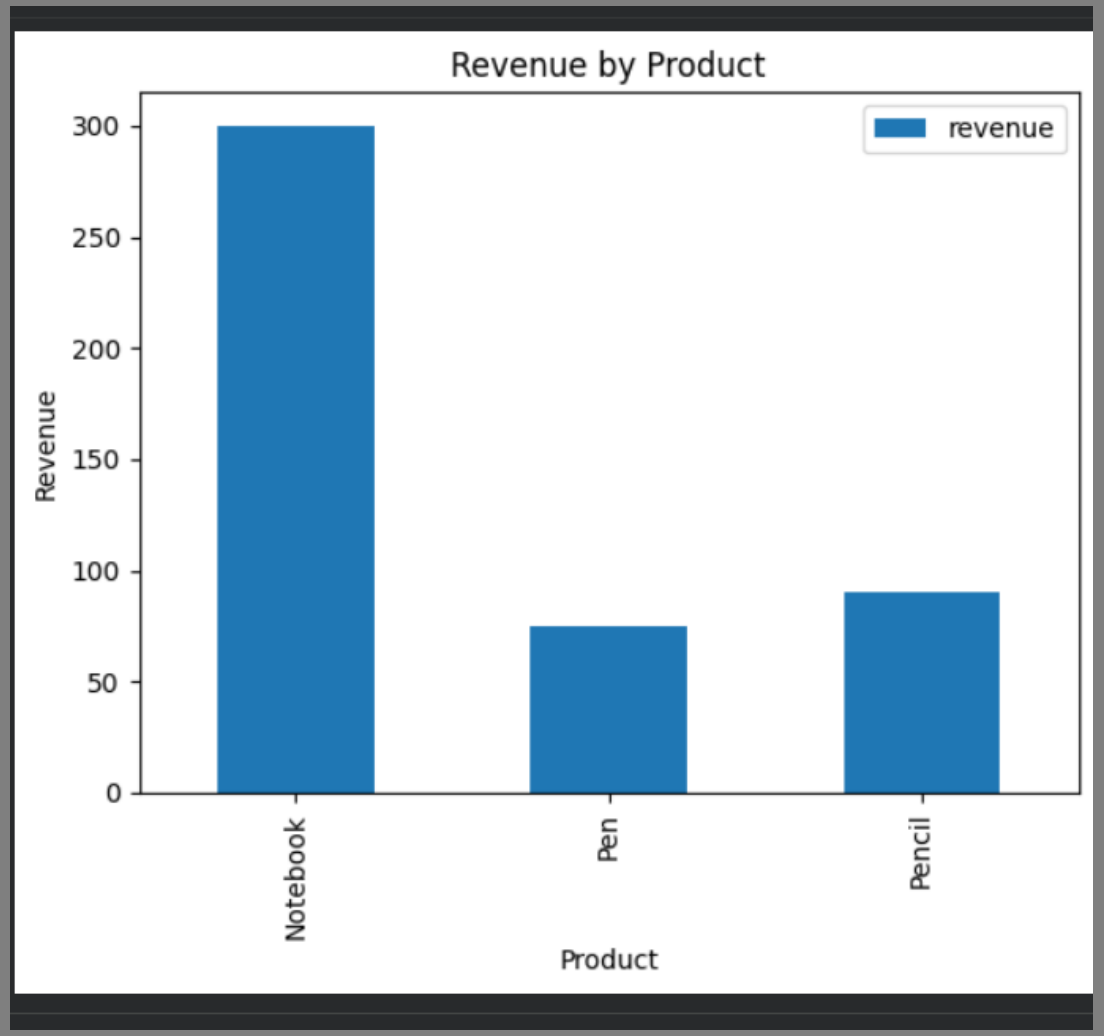
Pencil generated moderate revenue

Pen generated the lowest revenue

Insight:

Higher unit price significantly increased total revenue

SQL aggregation results validated through visualization



Data Validation – Sales Table

Purpose:

Verify that data was inserted correctly into the database

Ensure table structure is accurate before performing aggregation

SQL Query Used:

SELECT * FROM sales

Retrieved all rows from the sales table

Observation:

5 records successfully inserted

Columns: product, quantity, price

Data ready for further analysis

```python
# Check all data inside sales table
check_query = "SELECT * FROM sales"

df_check = pd.read_sql_query(check_query, conn)

print("All Sales Data:")
print(df_check)
```

```
All Sales Data:
    product  quantity  price
0       Pen        10    5.0
1    Pencil        20    3.0
2  Notebook        15   20.0
3       Pen         5    5.0
4    Pencil        10    3.0
```

- **Outcome:**
- Completed an end-to-end mini data analysis workflow
- Demonstrated ability to combine database querying with visualization
- **Tools Used:** Python | SQLite | Pandas | Matplotlib | Google Colab

THANKYOU